

DoTA 2 Hero Synergy Predictor

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The problem



DoTA 2 is a complex game with over 100 heroes.



Team cohesion and hero synergy are key factors in attaining victory.



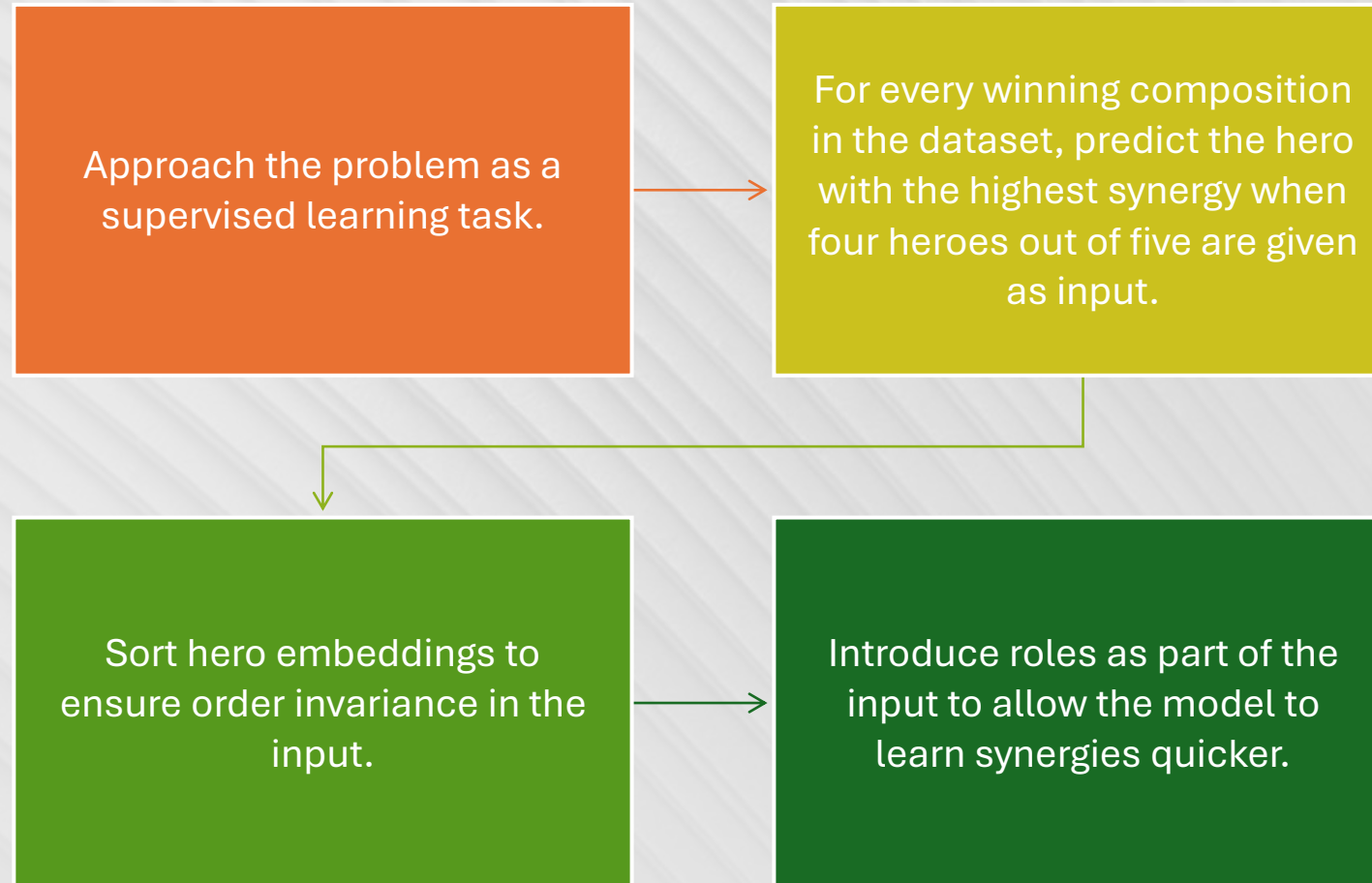
Not all heroes are equally compatible – some heroes are better together while others are anti-synergistic.



Our approach

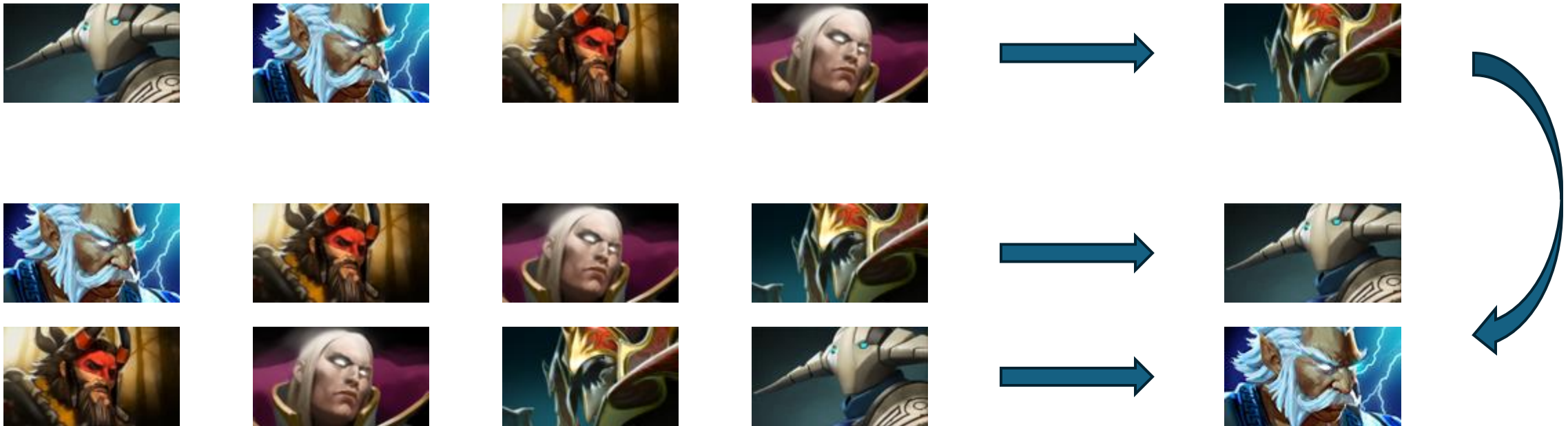
- Design an Artificial Neural Network with embeddings and attention layers to predict synergies between heroes.
- Aim to create a model that can consistently predict heroes that will fit well into a team.
- Leverage the model to analyse and discover new synergies that were previously not popular.

Implementation detail



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- Augment the dataset with rotated teams to allow the model to generalize the relations between heroes further.



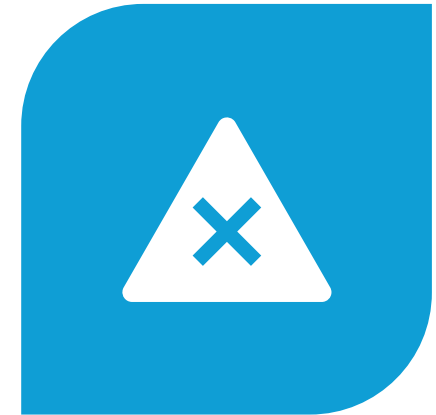
The results



THE MODEL ACHIEVED A REMARKABLE
ACCURACY OF 76%

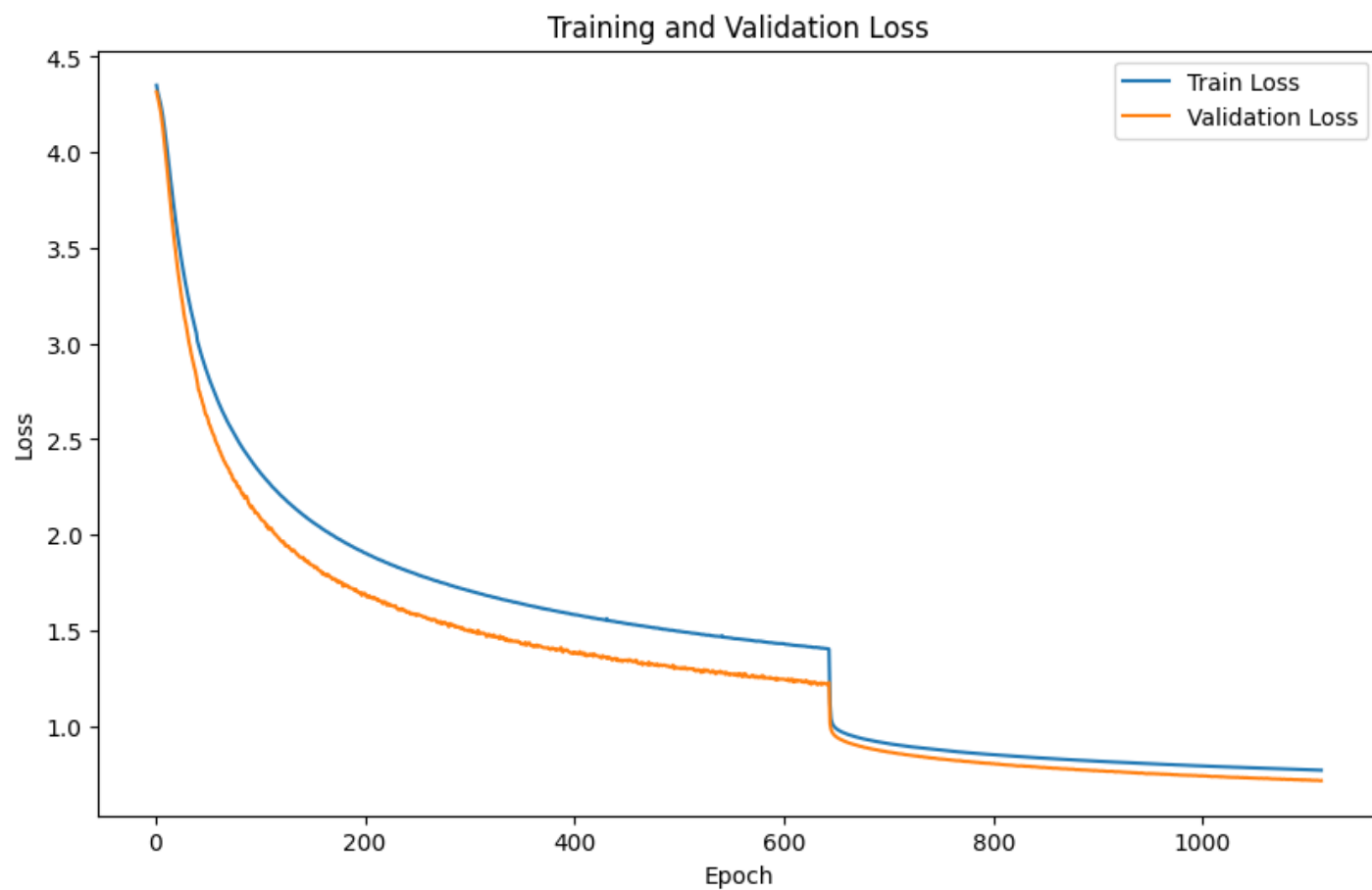


AUGMENTING THE DATASET IMPROVED THE
MODEL'S ABILITY TO GENERALIZE SYNERGIES –
EXCHANGING THE OUTPUT WITH THE INPUTS
YIELDED SIMILAR RESULTS.

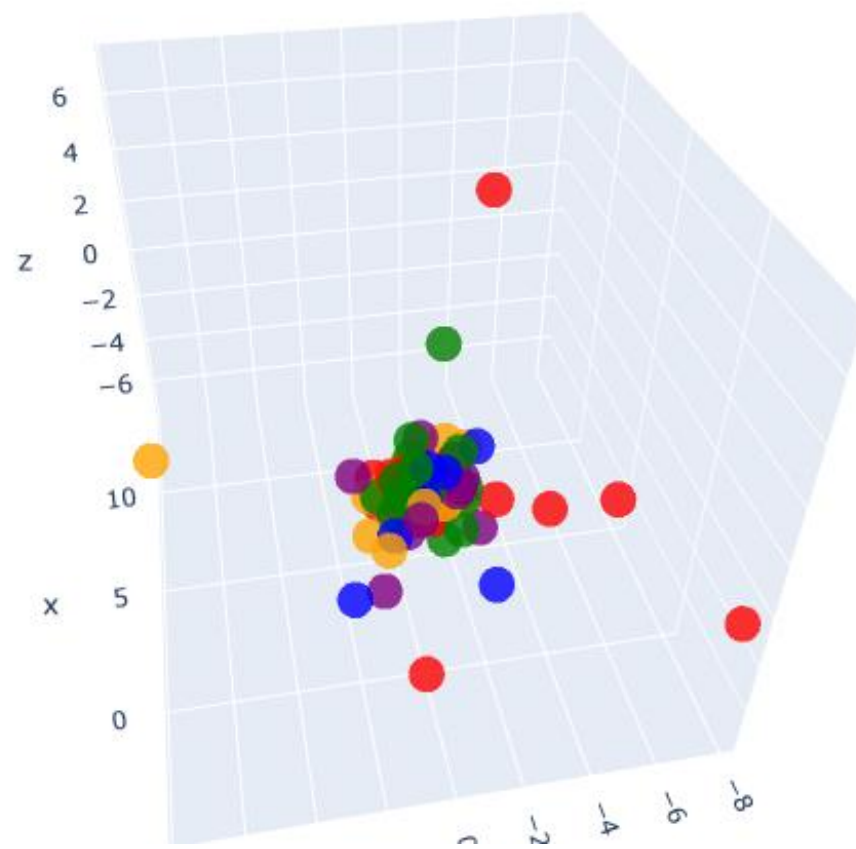


ORDER INVARIANCE ENSURES THAT THE
MODEL DOES NOT GIVE UNDUE BIAS TO THE
ORDER IN WHICH HEROES ARE INPUTTED.

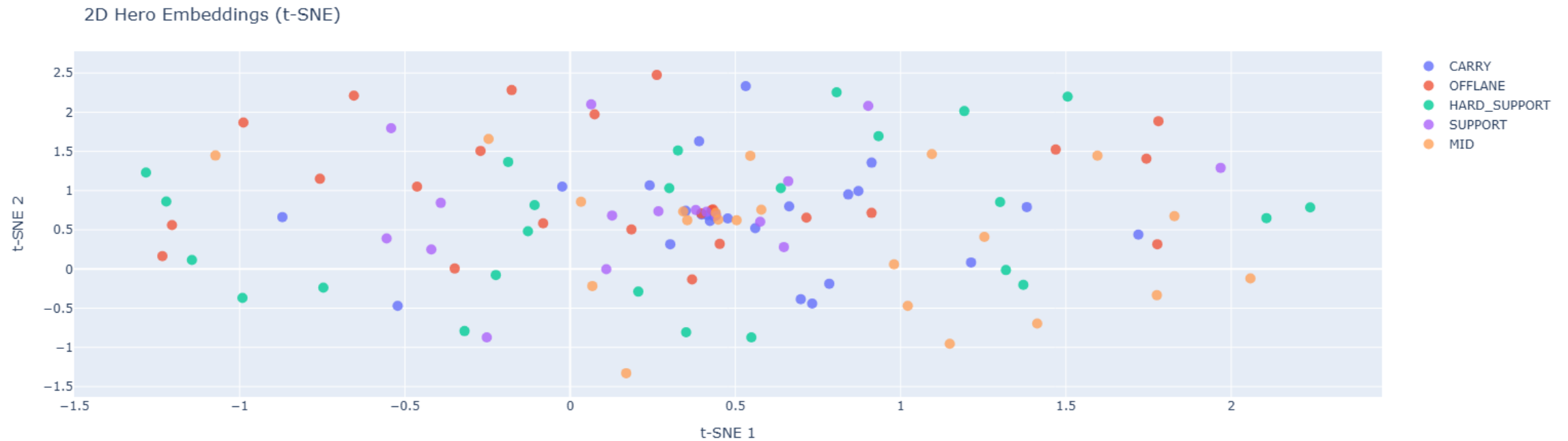
Training and validation loss curve



Hero embeddings in 3D space



Hero embeddings in 2D space



Hero synergies in a network graph



Conclusion

- Predicting team compositions and synergies is a much more feasible and pragmatic task compared to predicting match outcomes.
- Data augmentation significantly improved the generalization power of the model.
- Embeddings allowed us to express heroes in a higher dimensional space and visualize them

